# Design: Variational Transfer for Impedance-Overlap Schwarz on Non-Conforming Meshes

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## 1 Problem statement

The impedance-overlap transmission condition

$$\boldsymbol{t}_k - \boldsymbol{t}_j + \boldsymbol{Z}\left(\dot{\boldsymbol{u}}_k - \dot{\boldsymbol{u}}_j\right) + \alpha\left(\boldsymbol{u}_k - \boldsymbol{u}_j\right) = \boldsymbol{0} \qquad \text{on } \Gamma_k \tag{1}$$

is now exactly conservative on conforming meshes (0.08% PU-energy error over 1 ms with the consistent partner traction, matching the exact-DBC reference). On the non-conforming cantilever pair ($h = 8.47$ vs. 6.35 mm, 4:3) the net error also looks small (±1%), but the interface-work instrumentation shows that this is a cancellation of two error channels, each of which moves roughly *half the system energy per millisecond*:

| 1 ms, II, PU metric | conforming | non-conforming $h$ | non-conforming $h/2$ |
|---|---|---|---|
| dashpot (spurious) work | −42.6 J (2.8%) | −606.5 J (40%) | −598.5 J (40%) |
| traction-channel work | — | +677 J (45%) | +768 J (51%) |
| net PU-energy error | 1.5% → 0.08% | −0.75% (growth) | +1.5% (loss) |
| mean velocity jump (RMS) | 8.9 m/s | 40 m/s (growing 15→60) | 27 m/s |

Three facts drive the design. (i) The channels do *not* shrink under mesh refinement (dashpot work unchanged at $h/2$), so resolution cannot fix the interface; the pointwise-interpolation transfer feeds the condition an $O(1)$ error on mesh-scale content. (ii) The net error is the difference of two large channels and even flips sign between $h$ and $h/2$ — it is not robust, which is the production fragility. (iii) The velocity jumps grow through the run: each interface pass leaves mesh-scale residue that reverberates, a bounded version of the same feedback that makes DBC overlap coupling exponentially unstable.

The theoretical requirements are Gander–Halpern–Nataf's three conditions for discrete energy convergence of waveform-relaxation coupling on nonmatching grids: (i) Robin (not Dirichlet) transmission, (ii) consistent boundary-cell discretization of the transmission operator, (iii) an $L^2$-contractive intergrid transfer. The impedance-overlap BC satisfies (i); its weak (surface-mass-weighted) assembly satisfies (ii); pointwise interpolation violates (iii). Replacing the transfer with the variational ($L^2$-projection, known as mortar in the domain-decomposition literature) transfer onto the receiving trace space satisfies all three simultaneously. Note that weak-$L^2$ enforcement of *Dirichlet* overlap coupling did not remove the DBC instability — consistent with the theory, since DBC violates (i) regardless of the transfer; the combination proposed here is the first that meets all three conditions.

## 2 Operator inventory (existing machinery)

All operators needed for the first stage already exist:

| operator | code | status |
|---|---|---|
| $\boldsymbol{W}_k = \int_{\Gamma_k} \boldsymbol{N}_k \boldsymbol{N}_k^\top \mathrm{d}\Gamma$ | `get_square_projection_matrix` | already on the BC |
| $\boldsymbol{L}_{kj} = \int_{\Gamma_k} \boldsymbol{N}_k(\boldsymbol{x})\, \boldsymbol{N}_j^{\mathrm{vol}}(\boldsymbol{x})^\top \mathrm{d}\Gamma$ | `get_overlap_rectangular_projection_matrix` | exists (weak-DBC path); needs signature relaxed |
| $\boldsymbol{P}_{kj} = \boldsymbol{W}_k^{-1} \boldsymbol{L}_{kj}$ | weak-DBC `dirichlet_projector` | same construction |
| surface-to-surface $\boldsymbol{R}_{kj}$ | `get_rectangular_projection_matrix` (`project_point_to_side_set`) | exists (nonoverlap path); needed in stage 2 |
| partner one-sided weak flux | `build_consistent_traction_patch!`, `consistent_partner_traction` | exists (conforming); generalized in stage 2 |

$\boldsymbol{L}_{kj}$ is assembled by quadrature over $\Omega_k$'s boundary facets; at each quadrature point the partner *volume* element is located (`find_point_in_mesh`) and its shape functions evaluated — exactly the variational cross-mass for an overlap boundary, which is interior to the partner and therefore has no partner surface mesh.

Two structural properties come for free. First, $\boldsymbol{P}_{kj} = \boldsymbol{W}_k^{-1} \boldsymbol{L}_{kj}$ is the $L^2(\Gamma_k)$-orthogonal (Galerkin) projection onto $\Omega_k$'s trace space, hence non-expansive in $L^2$: GHN condition (iii) holds by construction, for any quadrature, because it is a projection. Second, partition of unity of the partner shape functions gives $\boldsymbol{L}_{kj}\boldsymbol{1} = \boldsymbol{W}_k \boldsymbol{1}$ for *any* quadrature rule, so constants (rigid translations) transfer exactly regardless of quadrature accuracy.

## 3 Design

### 3.1 Stage 1: variational projection of the transferred fields

Replace pointwise interpolation of the three partner fields with the variational projection, leaving everything downstream of the nodal values unchanged:

$$\hat{\dot{\boldsymbol{u}}}_j = \boldsymbol{P}_{kj}\, \dot{\boldsymbol{u}}_j, \qquad \hat{\boldsymbol{u}}_j = \boldsymbol{P}_{kj}\, \boldsymbol{u}_j, \qquad \hat{\boldsymbol{\sigma}}_j = \boldsymbol{P}_{kj}\, \boldsymbol{\sigma}_j^{\mathrm{rec}} \quad \text{(six Voigt components)}, \tag{2}$$

then form the traction $\hat{\boldsymbol{t}}_j = \hat{\boldsymbol{\sigma}}_j\, \boldsymbol{n}_k$ and the tensor-impedance term nodally exactly as the current code does. This is a drop-in change: `apply_bc_detail`'s per-node interpolation loop becomes three (or nine, per component) small dense matrix–vector products with a projector precomputed at setup. The P/S-split tensor $\boldsymbol{Z}(\boldsymbol{n})$ continues to act at the receiving nodes, so no re-derivation of the self terms or the tangent is needed; the variational crime of nodal application is unchanged from the current scheme and identical on both branches.

Implementation touch points:

- **YAML**: new BC option `transfer:` with values `pointwise` (current behavior, initial default) and `variational`.
- **Types**: field `variational_projector::Matrix{Float64}` ($n_k \times N_j$, dense; boundary node count times partner node count, small for realistic interfaces) on `SolidMechanicsImpedanceOverlapSchwarzB`

- **Setup** (`compute_impedance_overlap_schwarz_projectors!`): relax `get_overlap_rectangular_projectio` signature to accept the impedance BC (it only touches fields both BC types share), build $\boldsymbol{L}_{kj}$, factorize the existing $\boldsymbol{W}_k$ (Cholesky) once, store $\boldsymbol{P}_{kj}$. Rebuilt on BC swaps, like the consistent-traction patch.

- **Evaluation**: factor the partner-field sampling shared by `apply_bc_detail` and `report_impedance_interfac` into one helper that dispatches on the transfer mode, so the instrumentation always measures what the BC actually applies.

- **Precedence**: on node-aligned interfaces $\boldsymbol{L}_{kj}$ has a single unit entry per row and $\boldsymbol{P}_{kj}$ reduces to the identity; the consistent-traction patch (`partner traction:  auto`) continues to take priority for the traction, and the kinematic projection becomes a no-op. The conforming limit is therefore automatic.

Quadrature note: the integrand of $\boldsymbol{L}_{kj}$ is only piecewise smooth on $\Gamma_k$ (partner elements change within a receiving facet), so the default facet rule carries a consistency error. Because contractivity and constant-transfer hold for any rule, this error degrades accuracy, not stability. The cheap knob is an elevated or subdivided facet rule for the $\boldsymbol{L}_{kj}$ assembly only (setup cost); a true common-refinement (2D polygon overlay on the flat interface) is the exact alternative and is deferred until measurements justify it.

What stage 1 answers: how much of the 600 J churn is the *transfer* (interpolation aliasing of mesh-scale content) versus the *recovered-stress traction*. The conforming ladder bounds what recovery-quality traction can achieve ($\approx$1.5%-class); if variational transfer restores $O(h)$ scaling of the channels, refinement plus stage 2 closes the rest.

### 3.2 Stage 2: consistent traction across the interface

The conforming analysis showed the traction channel is the dominant residual once transfer is clean, and that nodal recovery cannot represent the traction carried by mesh-scale content. The consistent (one-sided partial assembly) traction requires a partner facet surface bounding a one-sided element patch; on a non-conforming interface $\Gamma_k$ cuts through partner elements, so the patch does not exist on $\Gamma_k$ itself. Two generalizations, in order of increasing cost:

**2a: offset facet surface + surface variational.** Take $\tilde{\Gamma}_j$ = the partner facet surface immediately exterior to $\Gamma_k$ (the boundary between the first exterior element layer and the rest; the patch-construction logic already classifies these elements). The consistent flux is exact on $\tilde{\Gamma}_j$ by the same partial assembly as the conforming case, tested with partner surface functions. Transfer the resulting weak traction to $\Gamma_k$'s trace space with the surface-to-surface variational transfer (`get_rectangular_projection_matrix` generalized to accept the internal surface in place of a side set, or an equivalent assembled operator), i.e. $\boldsymbol{W}_k\hat{\boldsymbol{t}}_j = \boldsymbol{R}_{k\tilde{j}}\boldsymbol{W}_{\tilde{j}}^{-1}\lambda_{\tilde{j}}$. The surface offset introduces an $O(h_j)$ modeling error ($\tilde{\Gamma}_j$ is within one partner element of $\Gamma_k$), but it acts on the *exact discrete flux* rather than on recovered stress; whether that trade wins is an experiment, not a theorem — measure before committing further.

**2b: cut-cell partial assembly (flat interfaces).** For a planar $\Gamma_k$ (the common production configuration), integrate the partner's momentum residual over the plane-clipped parts of the cut elements, testing with $\Omega_k$'s surface functions extended constant along the interface normal. This reproduces the conforming construction exactly in the limit and removes the offset error of 2a, at the cost of new machinery (hexahedron–half-space clipping and quadrature on the clipped polyhedra). Deferred unless 2a measures short of the target.

### 3.3 Validation and targets

The metric is the instrumentation's channel decomposition, not the net energy: success is driving the churn itself down, since the current scheme already fakes a small net by cancellation.

1. **Conforming regression**: variational transfer on the node-aligned pair must reproduce the pointwise results exactly ($\boldsymbol{P} = \boldsymbol{I}$); the consistent-traction 0.08% result must be unaffected. Extend test 103/104 asserts with the projector identity check.
2. **Non-conforming, stage 1**: dashpot work $\ll 600\,\mathrm{J}$ at $h$, and restored $O(h)$ decay at $h/2$; velocity-jump growth curve flattened. Expected end state $\approx$ conforming recovered-stress class (few percent), now robust rather than cancellation-balanced.
3. **Non-conforming, stage 2**: traction channel reduced toward the consistent-traction class; net PU error $\lesssim 1\%$ with both channels individually small; sign and magnitude stable under refinement.
4. **Stability margin**: rerun the impedance-scale sweep — with contractive transfer the growth branch above scale 1 should soften, and the DBC-limit blow-up should set in later; this is the direct measure of production robustness.
5. **Cost**: setup $O(n_{\text{facets}} \times n_{\text{qp}})$ point locations (same as the weak-DBC path); per-iteration cost *decreases* (dense mat-vec replaces per-node interpolation loops; no recovery solve once stage 2 lands).

## 4 Stage 1 measurements (2026-07-03)

Stage 1 was implemented (`transfer: variational`, commit-level detail in the repository) and measured on the instrumented 1 ms non-conforming benchmark:

| 1 ms, II, PU metric | pointwise $h$ | pointwise $h/2$ | variational $h$ | variational $h/2$ |
|---|---|---|---|---|
| dashpot work [J] | $-606.5$ | $-598.5$ | $-726.1$ | $-483.9$ |
| traction-channel work [J] | $+677$ | $+768$ | $+880$ | $+700$ |
| net PU-energy error | $-0.75\%$ | $+1.5\%$ | $+6.2\%$ (growth) | $+0.7\%$ (growth) |
| mean velocity jump [m/s] | 40.0 | 27.0 | 37.8 | 28.0 |

Three conclusions. First, the variational transfer restores an $h$-dependence that pointwise interpolation lacks (dashpot churn $726 \to 484\,\mathrm{J}$ under refinement, versus unchanged), but the decay is slow and the channel magnitudes remain in the hundreds of joules — the error class is unchanged. Second, at the coarse resolution the net error *worsens* (6.2% growth versus 0.75%): under pointwise interpolation the kinematic-transfer error and the recovered-traction error come from the same interpolant and partially cancel inside the transmission condition; projecting the kinematics decorrelates them, and the interface becomes net-injecting. Third, the traction channel is transfer-dependent and did not improve (+880 J at $h$): the recovered-stress traction remains the dominant polluter.

The design consequence: **stage 2 is not optional**. The kinematic variational transfer should land *together* with a variational (consistent) traction — shipping stage 1 alone is counterproductive at production resolutions.

**Quadrature experiment.** Elevating the facet quadrature of $\boldsymbol{L}_{kj}$ (a $4 \times 4$ sub-cell rule via the new `transfer quadrature subdivisions` option, sub-cells of 2.1 mm against 6.35 mm partner elements) separates the quadrature consistency error from the traction-channel error at $h$: the net error improves from +6.2% to +2.4% growth — so the default rule's consistency error was a material contributor to the stage-1 net degradation and the subdivided rule should be used whenever the facet-size ratio is coarse — while the channel magnitudes grow (dashpot $-726 \to -857$ J, traction $+880 \to +941$ J): the more faithful projection captures more of the partner's mesh-scale content, which the dashpot must then dissipate. Accurate transfer quadrature therefore does not change the conclusion: the recovered-stress traction channel dominates, and stage 2 remains the required fix.

## 5 Stage 2a measurements (2026-07-03)

Stage 2a was implemented (offset-surface consistent traction with the surface-to-surface transfer $\boldsymbol{T} = \boldsymbol{R}\tilde{\boldsymbol{W}}^{-1}$) and verified against a manufactured uniform uniaxial-stress state: the transferred weak traction reproduces $\boldsymbol{W}(\sigma \boldsymbol{n})$ to $10^{-11}$ relative error on the non-conforming pair, confirming the construction. (A uniform uniaxial-*strain* state fails the same test by design: its lateral confinement tractions are correctly picked up by the one-sided assembly on the patch's lateral boundary strips.) The instrumented benchmark, however, is decisively negative, completing a fidelity ladder at $h$:

| 1 ms, II, PU metric, $h$ | dashpot work | net PU error |
|---|---|---|
| pointwise + recovered stress | −607 J | −0.75% |
| variational + recovered stress | −726 J | +6.2% (growth) |
| variational, subdivided + recovered stress | −857 J | +2.4% (growth) |
| variational, subdivided + consistent traction (2a) | −1129 J | +9.3% (growth) |

Each fidelity increase makes the interface churn *worse*. The reading: on non-matching grids the round trip cannot be the identity; whatever content cannot be represented across the interface must be dissipated, or it aliases and reverberates. Nodal stress recovery had been acting as a beneficial low-pass filter; the exact discrete flux transmits the partner's full mesh-scale content into a trace space that cannot carry it, and the feedback loop (growing velocity jumps) is pumped harder. Fidelity without a mechanism that removes non-transferable content is anti-stabilizing.

A second structural defect is now visible: the traction arrives through $\boldsymbol{T}$ (built on the offset surface) while the velocity arrives through $\boldsymbol{P}$ (built from volume interpolation) — two different filters. The impedance condition has characteristic structure, $t_k - Z\dot{u}_k = -(t_j + Z\dot{u}_j)$ up to signs: for outgoing waves the combination $g_j = \boldsymbol{t}_j + \boldsymbol{Z}\dot{\boldsymbol{u}}_j + \alpha \boldsymbol{u}_j$ is smooth even when $\boldsymbol{t}_j$ and $\dot{\boldsymbol{u}}_j$ separately carry mesh-scale standing content, because that content cancels *within* $g_j$. Transferring the two pieces through different operators destroys the cancellation at exactly the mesh scale where it matters.

**Proposed stage 2c: single-operator characteristic transfer.** Assemble the partner's full Robin datum on the offset surface — $\tilde{\boldsymbol{g}} = -\tilde{\lambda} + \tilde{\boldsymbol{W}}\,(\boldsymbol{Z}\,\dot{\boldsymbol{u}}_j + \alpha\,\boldsymbol{u}_j)\big|_{\tilde{\Gamma}}$, using the partner's own nodal trace values on $\tilde{\Gamma}$ (no interpolation at all) — and transfer it with the single operator $\boldsymbol{T}$. The mesh-scale cancellation then happens on the partner side, before transfer. Implementation cost is small: keep $\tilde{\boldsymbol{W}}$ and the $\tilde{\Gamma}$ node list in the patch, and bypass $\boldsymbol{P}$ in the right-hand side. This was implemented and measured; see the following section — the hypothesis did not survive.

## 6 Stage 2c measurements and revised conclusions (2026-07-03)

Stage 2c was implemented (single-operator characteristic transfer of the complete Robin datum, assembled on $\tilde{\Gamma}$ from the partner's own nodal trace values) and benchmarked identically. The result is statistically indistinguishable from stage 2a: dashpot work $-1084$ vs $-1129$ J, net $+9.8$ vs $+9.3\%$ growth, the same velocity-jump growth. The characteristic-cancellation hypothesis is refuted: the operator pairing between traction and velocity is not the driver.

The complete five-variant ladder at $h$ now reads: pointwise + recovered $-607$ J / $-0.75\%$; variational + recovered $-726$ J / $+6.2\%$; + subdivided quadrature $-857$ J / $+2.4\%$; + consistent traction (2a) $-1129$ J / $+9.3\%$; characteristic (2c) $-1084$ J / $+9.8\%$. Every fidelity increase pumps the churn. The conclusion is structural, not implementational (both stages pass the uniform-stress patch test to $10^{-11}$): **on non-matching grids, content that cannot be represented across the interface must be dissipated, and smoothing the transferred data is beneficial — the recovered-stress low-pass filter was doing load-bearing stabilization work.** The parameter-free pointwise + recovered-stress scheme (with the partner traction term and consistent recovery) is empirically the best of the five variants at production resolution.

**Where this leaves the design.** (i) For production non-conforming runs, keep `transfer: pointwise` with recovered stress as the default; the variational machinery remains valuable for its conforming-limit exactness, the quadrature knob, and as diagnostic infrastructure. (ii) The principled next idea is a *content-aware absorbing term*: dissipate exactly the non-transferable component by penalizing $(\boldsymbol{I} - \Pi)\,\dot{\boldsymbol{u}}_k$ with the characteristic impedance, where $\Pi$ is the round-trip transfer operator — absorb what cannot cross, transmit what can. This turns the accidental stabilization of the recovery filter into a designed mechanism with a conforming limit of zero. (iii) A cheap discriminating diagnostic first: rerun the benchmark with a dissipative integrator (Newmark $\gamma > 1/2$ or HHT-$\alpha$) to confirm the churn is carried by the high-frequency band that subdomain-level damping removes.

## 7 Content-aware absorption and the closing diagnostic (2026-07-03)

The content-aware absorbing term was implemented (BC option `content aware absorption`): an additional dissipative self term $\boldsymbol{W}\boldsymbol{Z}(\boldsymbol{I} - \Pi)\,\dot{\boldsymbol{u}}_k$ with its Newmark tangent, where $\Pi$ is the $\boldsymbol{W}$-orthogonal projection onto the span of the variational projector's columns — the *transferable* space at the boundary. The filter has exactly the designed structure: it annihilates constants ($\boldsymbol{P}\boldsymbol{1} = \boldsymbol{1}$ puts rigid motions in the span), vanishes identically on node-aligned interfaces and on the coarse side of the study pair (everything a coarse trace holds is representable by the fine partner), is nonzero only on the fine side ($\|\boldsymbol{F}\| = 0.44$), and is provably dissipative ($v^\top \boldsymbol{W}\boldsymbol{F}\,v \geq 0$ per scalar channel; asserted in test 105).

**The measured effect is null**: the absorber drains 4.6 J over 1 ms while the channels and the net remain identical to the pointwise baseline. Because the operator is verified, the null is informative: *the reverberating content lives inside the transferable space.* Both grids can represent the mesh-scale content that churns; they disagree about it because their discrete dispersion relations differ. No transfer-side construction can remove a dispersion disagreement, by construction — which retroactively explains the entire fidelity ladder.

The closing diagnostic confirms the mechanism: rerunning the pointwise baseline with a dissipative integrator (Newmark $\gamma = 0.6$, $\beta = 0.3025$) collapses the churn from $-607$ to $-231$ J and, decisively, the velocity jumps *decay* ($11.8 \to 5.6$ m/s between run halves) instead of growing ($33 \to 47$): the reverberation loop is dead. The churn is carried by the high-frequency band that subdomain-level damping removes. ($\gamma = 0.6$ is a blunt instrument — it also removed 87% of the physical energy on this broadband benchmark; production should use mild HHT-$\alpha$-class damping concentrated near the grid Nyquist.)

**Final recommendations.** (i) **Conforming interfaces**: consistent traction — 0.08%, exact-DBC class. (ii) **Non-conforming production runs**: pointwise transfer + recovered-stress traction with the physical impedance (parameter-free), paired with mild high-frequency integrator damping; quantify the damping trade-off with a dissipation sweep. (iii) The variational machinery (projector, subdivided quadrature, offset-surface flux, characteristic transfer, content filter) is retained: it provides the conforming-limit exactness, the quadrature control, the instrumentation-grade diagnostics that produced these conclusions, and patch-test-verified building blocks for future transmission-condition work. (iv) The OSM-level lesson: the non-conforming overlap interface error is dominated by the discrete dispersion disagreement of high-frequency content between the subdomain discretizations — a property of the grids, not of the transmission condition. The impedance dashpot is the correct interface-level treatment and is already in place; the remaining energy budget belongs to the time integrator.

## 8 HHT-$\alpha$ and the damping trade-off (2026-07-03)

HHT-$\alpha$ numerical dissipation was added to the Newmark integrator (`HHT alpha:` $\bar{\alpha} \in [0, 1/3]$; internal force assembled at the blended state $(1-\bar{\alpha})u_{n+1} + \bar{\alpha}u_n$, $\gamma = 1/2 + \bar{\alpha}$ and $\beta = (1+\bar{\alpha})^2/4$ derived, $\rho_\infty = (1-\bar{\alpha})/(1+\bar{\alpha})$) and swept on both mesh pairs at 1 ms, $\Delta t = 10^{-6}$ s:

| $\bar{\alpha}$ | 0 | 0.02 | 0.05 | 0.1 | 0.2 |
|---|---|---|---|---|---|
| non-conforming $E/E_0$ | 1.007 | 0.952 | 0.892 | 0.833 | 0.781 |
| conforming $E/E_0$ (damping cost) | 0.999 | 0.969 | 0.938 | 0.905 | 0.873 |
| interface-specific extra loss | ±1% | 1.7% | 4.9% | 8.0% | 10.5% |
| dashpot churn [J] | −607 | −595 | −580 | −561 | −540 |
| jump growth (halves, m/s) | $33 \to 47$ | $32 \to 44$ | $31 \to 41$ | $30 \to 37$ | $29 \to 32$ |

Three findings. First, mild HHT barely touches the churn even though $\bar{\alpha} = 0.1$ has the *same* $\rho_\infty = 0.818$ as the Newmark $\gamma = 0.6$ diagnostic that killed the loop: at $\Delta t = 10^{-6}$ s the mesh-frequency band sits at $\omega\Delta t \approx 0.3$–$0.7$, which the time step resolves and HHT (by design) preserves, while Newmark-$\gamma$'s broadband first-order damping reaches it — and the physics with it (87% loss vs 17% at the same $\rho_\infty$). Second, pushing the band into HHT's damped range with $\Delta t = 4 \times 10^{-6}$ s does not help *on this benchmark*: the conforming cost balloons to 31%, because the parabolic release has **no scale separation** — its physical content extends to the grid scale, so any damping strategy bills real energy. Third, the interface-specific extra loss *grows* with damping (1.7 to 10.5%): the interface keeps regenerating band content that the integrator then removes.

**Reading.** Damping converts the knife-edge ±1% (sign-indefinite, growth-capable) into monotone, bounded decay — robustness — at a cost proportional to how much of the problem's energy lives in the reverberating band. On this benchmark that fraction is large by construction; on production problems with scale separation (smooth loading, mesh resolving the physics with margin), $\bar{\alpha} \approx 0.02$–$0.05$ with a physics-sized $\Delta t$ buys the robustness at near-zero cost. Demonstrating that quantitatively needs a smooth-content benchmark — the natural companion to the bimaterial test in the open items.

# 9 Summary

Stage 1 is a low-risk, high-information change that reuses the square ($\boldsymbol{W}_k$) and rectangular ($\boldsymbol{L}_{kj}$) projection machinery verbatim: variationally project $\dot{\boldsymbol{u}}_j$, $\boldsymbol{u}_j$, and $\boldsymbol{\sigma}_j^{\mathrm{rec}}$ onto the receiving trace space and keep the rest of the impedance pipeline unchanged. The conforming limit and the consistent-traction path are preserved automatically. The stage-1 measurements show that contractive kinematics alone do not reduce the interface churn — they restore $h$-decay but break the error cancellation that kept the pointwise scheme's net small — so the variational transfer and the stage-2 variational traction must ship together, with the channel decomposition (not the net energy) as the acceptance metric.